

FAME PI meeting

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- PhD in mathematics at Aalto Uni. 2017
- Postdoc at UNSW Sydney 2018–2020
- Postdoc at LUT 2020–2022
- Postdoc at FU Berlin 2022–2025
 - Substitute professor at Uni. Potsdam 2024
- Associate professor at LUT 2025–

High-dimensional numerical integration

$$\int_{[0,1]^s} f(\mathbf{y}) \, d\mathbf{y} \approx \sum_{i=1}^n w_i f(\mathbf{t}_i)$$

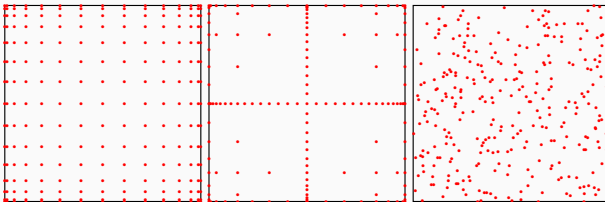


Figure 1: Tensor product grid, sparse grid, Monte Carlo nodes (not QMC rules)

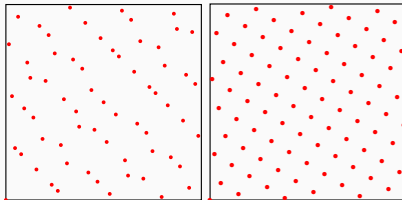


Figure 2: Sobol' points, lattice rule (examples of QMC rules)

Uncertainty in groundwater flow

Risk analysis of radwaste disposal or CO₂ sequestration.

Darcy's law

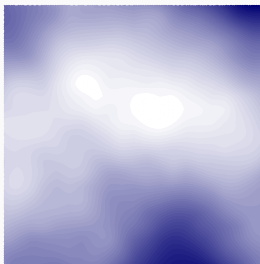
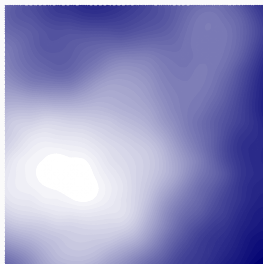
$$\mathbf{q}(\mathbf{x}) + \mathbf{a}(\mathbf{x})\nabla p(\mathbf{x}) = \mathbf{f}(\mathbf{x})$$

mass conservation law

$$\nabla \cdot \mathbf{q}(\mathbf{x}) = 0$$

in $D \subset \mathbb{R}^d$, $d \in \{1, 2, 3\}$

together with boundary conditions

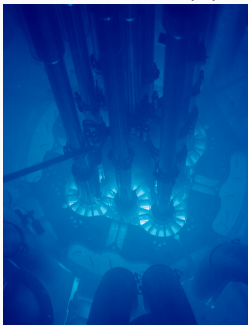


Uncertainty in $\mathbf{a}(\mathbf{x}, \omega)$ leads to uncertainty in $\mathbf{q}(\mathbf{x}, \omega)$ and $p(\mathbf{x}, \omega)$

Criticality problem for nuclear reactors

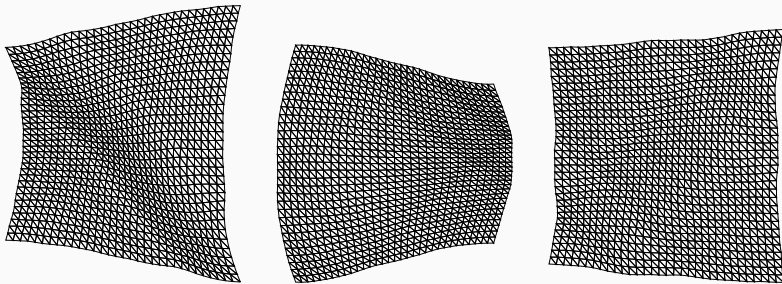
$$-\nabla \cdot \underbrace{\left(a(\mathbf{x}) \nabla u(\mathbf{x}) \right)}_{\text{diffusion}} + \underbrace{b(\mathbf{x})}_{\text{absorption}} u(\mathbf{x}) = \lambda \underbrace{c(\mathbf{x})}_{\text{fission}} u(\mathbf{x})$$

- The smallest eigenvalue $\lambda_1 \in \mathbb{R}$ measures **criticality** of a reactor.
- Eigenfunction $u_1(\mathbf{x})$ is the **neutron flux** at the point $\mathbf{x}$.



- $\lambda_1 \approx 1 \Rightarrow$ operating efficiently
- $\lambda_1 > 1 \Rightarrow$ not self-sustaining
- $\lambda_1 < 1 \Rightarrow$ supercritical

Source: Argonne National
Laboratory on Flickr

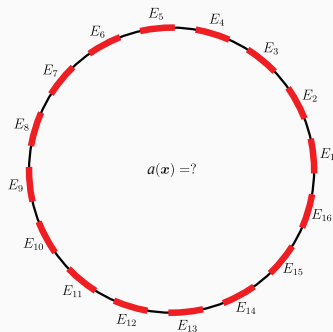


Three realizations of a random domain

Electrical impedance tomography

Use measurements of current and voltage collected at electrodes covering part of the boundary to infer the interior conductivity of an object/body.

$$\begin{cases} \nabla \cdot (a \nabla u) = 0 & \text{in } D, \\ a \frac{\partial u}{\partial \mathbf{n}} = 0 & \text{on } \partial D \setminus \bigcup_{k=1}^L \overline{E_k}, \\ u + z_k a \frac{\partial u}{\partial \mathbf{n}} = \mathcal{U}_k & \text{on } E_k, \quad k \in \{1, \dots, L\}, \\ \int_{E_k} a \frac{\partial u}{\partial \mathbf{n}} \, dS = I_k, & k \in \{1, \dots, L\}, \end{cases}$$



- We are always looking for new problems where we can apply QMC
 - Do you have a problem where QMC might be useful?
- Recent trends: function approximation using QMC point sets, training neural networks over QMC point sets, superconvergence properties of smooth functions, QMC theory for nonsmooth functions, high-dimensional double integrals (design of experiments),...

Thank you for your attention!